

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	29 September 2026
Team ID	SWTID-2026-4968
Project Name	Script - controlled ACL - Restrict Record Access Based On Field Value
Maximum Marks	4 Marks

Technical Architecture:

The deliverable shall include the architectural diagram and the information as per Table-1 & Table-2.

Example: Script-controlled ACL for restricting record access based on field value

Reference: Project-specific access-control implementation using ACL conditions and scripts.

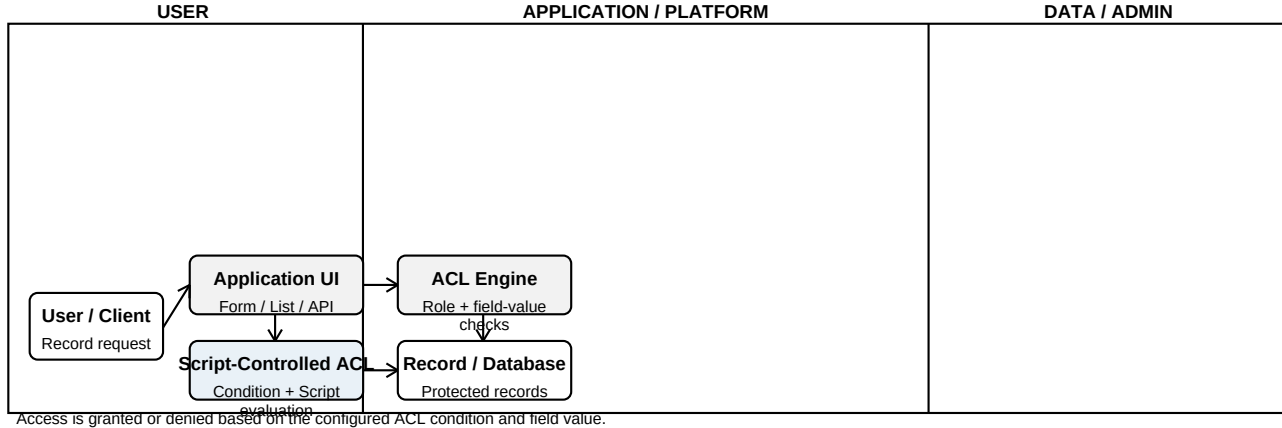


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	How user interacts with the application to view, create or update records.	HTML, CSS, JavaScript / ServiceNow UI

Table-1 : Components & Technologies (continued)

S.No	Component	Description	Technology
2.	Application Logic-1	Checks the requested operation and identifies the record and field value involved in the access decision.	JavaScript / Server-side Script
3.	Application Logic-2	Evaluates the ACL condition and executes the configured script to determine whether access is allowed.	ACL Condition + Script
4.	Application Logic-3	Applies role and field-value rules before permitting read, write, create or delete operations.	ACL Engine / Role-based Access Control
5.	Database	Stores application records and the field values used by the access-control rule.	Relational / Platform Database
6.	Cloud Database	Database service used when the application is deployed on a cloud/platform environment.	Platform Cloud Database
7.	File Storage	Stores configuration or supporting files required by the application, where applicable.	Platform Storage / Local Filesystem
8.	External API-1	Optional external interface used when an access decision requires data from another service.	REST API / Web Service
9.	External API-2	Optional external interface for retrieving additional business or identity information.	REST API / Web Service
10.	Machine Learning Model	Not required for the field-value ACL decision; access is determined by configured rules and scripts.	Not Applicable
11.	Infrastructure (Server / Cloud)	Hosts the application, ACL logic and database services used for secure record access.	Application Platform / Cloud or Local Server

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	List the open-source frameworks used by the application, if any.	JavaScript / Platform APIs
2.	Security	Restricts record access through ACL rules, roles, conditions and field-value checks.	ACL, Role-based Access Control, Script Validation
3.	Scalable Architecture	Separates UI, access-control logic and data services so rules can be maintained as application usage grows.	Layered Architecture / Service-based Platform
4.	Availability	Access-control evaluation is performed as part of the record request so unauthorized records are not exposed.	Platform Services / Distributed Deployment

Table-2: Application Characteristics (continued)

S.No	Characteristics	Description	Technology
5.	Performance	Design consideration for evaluating ACL rules efficiently, avoiding unnecessary database operations and keeping field-value checks focused.	Server-side Script / Indexed Fields / Platform Query Optimization

References:

- 1. Project-specific ACL configuration and script-controlled record access design.
- 2. Template structure supplied for Project Design Phase-II – Technology Stack (Architecture & Stack).